

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Tuesday

SET 10: Survey Responses Analysis

Aim

To analyze survey responses using **grouped bar chart, stacked bar chart, table, and Tableau dashboard** for interactive exploration of survey data.

Dataset

Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Q1. Grouped Bar Chart

Question

Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

R Program

```
# Question 1 responses

q1 <- c("A", "B", "C")

# Count each response

q1_count <- table(q1)

# Grouped bar chart

barplot(

  q1_count,

  beside = TRUE,

  names.arg = names(q1_count),

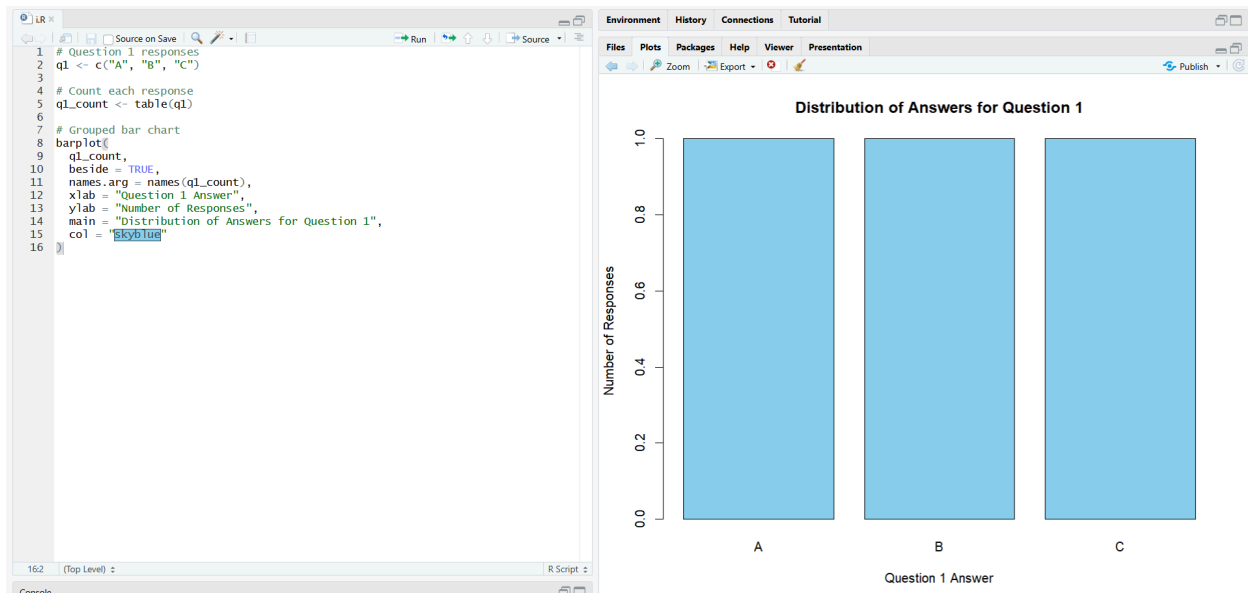
  xlab = "Question 1 Answer",

  ylab = "Number of Responses",

  main = "Distribution of Answers for Question 1",

  col = "skyblue"
```

Output



Observation

The three respondents selected **A, B, and C once each** for Question 1. Therefore, all three answer options have an equal frequency.

Q2. Stacked Bar Chart — Overall Response Distribution

Question

Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

R Program

```
# Survey responses

q1 <- c("A", "B", "C")

q2 <- c("B", "A", "A")

q3 <- c("C", "D", "B")

# Count responses for each question

q1_count <- table(factor(q1, levels = c("A", "B", "C", "D")))

q2_count <- table(factor(q2, levels = c("A", "B", "C", "D")))

q3_count <- table(factor(q3, levels = c("A", "B", "C", "D")))

# Combine counts

response_table <- rbind(

  "Question 1" = q1_count,

  "Question 2" = q2_count,

  "Question 3" = q3_count

)
```

```
# Stacked bar chart

barplot(

  response_table,

  beside = FALSE,

  col = c("skyblue", "lightgreen", "orange", "pink"),

  names.arg = c("Question 1", "Question 2", "Question 3"),

  xlab = "Questions",

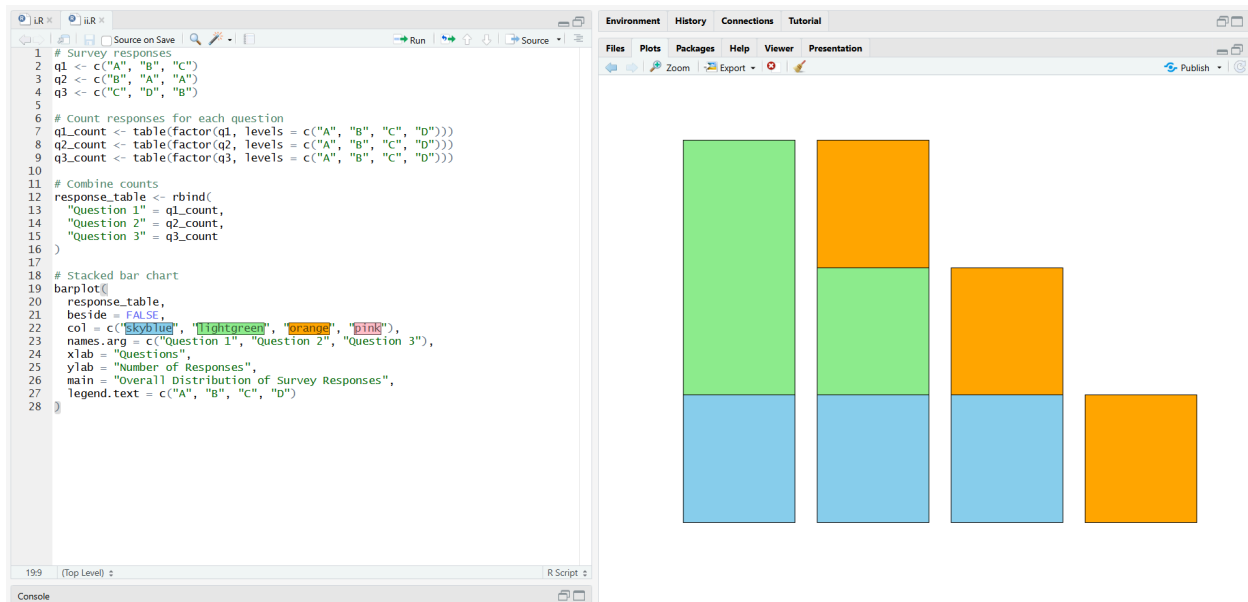
  ylab = "Number of Responses",

  main = "Overall Distribution of Survey Responses",

  legend.text = c("A", "B", "C", "D")

)
```

Output



Observation

- Question 1 has responses **A, B, C**.
- Question 2 has **A twice** and **B once**.
- Question 3 has **B, C, and D once each**.
- Each question has a total of **3 responses**.

Q3. Survey Response Table

Question

Build a table to show the survey response data for each survey. Label the table elements.

R Program

```
survey_table <- data.frame(  
  
  Survey_ID = c(1, 2, 3),  
  
  Question_1 = c("A", "B", "C"),  
  
  Question_2 = c("B", "A", "A"),  
  
  Question_3 = c("C", "D", "B")  
  
)  
  
print(survey_table)
```

Output

The screenshot shows an RStudio interface with three tabs at the top: 'i.R', 'ii.R', and 'iii.R'. The 'iii.R' tab is active, displaying the following R code:

```
1 survey_table <- data.frame(  
2   Survey_ID = c(1, 2, 3),  
3   Question_1 = c("A", "B", "C"),  
4   Question_2 = c("B", "A", "A"),  
5   Question_3 = c("C", "D", "B")  
6 )  
7  
8 print(survey_table)
```

Below the code editor is a status bar showing '8:20', '(Top Level)', and 'R Script'. At the bottom is a console window with tabs for 'Console', 'Terminal', and 'Background Jobs'. The 'Console' tab is active, showing the execution of the code and the resulting data frame:

```
> survey_table <- data.frame(  
+   Survey_ID = c(1, 2, 3),  
+   Question_1 = c("A", "B", "C"),  
+   Question_2 = c("B", "A", "A"),  
+   Question_3 = c("C", "D", "B")  
+ )  
>  
> print(survey_table)  
  Survey_ID Question_1 Question_2 Question_3  
1         1         A         B         C  
2         2         B         A         D  
3         3         C         A         B  
> |
```

Observation

The table successfully displays the responses given by each survey participant for all three questions.

Q4. Tableau Dashboard

Question

Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and table for interactive exploration of survey responses.

Tableau Procedure

Step 1 — Prepare the Data

Create an Excel/CSV file containing:

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Step 2 — Import into Tableau

1. Open **Tableau**.
2. Select **Microsoft Excel** or **Text File**, depending on your file.
3. Import the survey dataset.

Step 3 — Create Question 1 Bar Chart

- Drag **Question 1** → Columns.
- Drag **Survey ID** → Rows.
- Change aggregation to **Count**.
- Select **Bar**.
- Add a suitable title.

Title: Distribution of Question 1 Answers

Step 4 — Create Stacked Bar Chart

For the stacked chart, create a structure containing:

- Question
- Response
- Survey ID

Then:

- Drag **Question** → Columns.
- Drag **Survey ID** → Rows and use **Count**.
- Drag **Response** → Color.
- Keep the bars **stacked**.

Title: Overall Survey Response Distribution

Step 5 — Create Survey Table

Add:

- Survey ID
- Question 1
- Question 2
- Question 3

to the worksheet.

Step 6 — Create Dashboard

1. Select **New Dashboard**.
2. Add the Question 1 bar chart.
3. Add the stacked bar chart.
4. Add the survey response table.
5. Add filters if required.
6. Arrange and format the dashboard.

Expected Dashboard

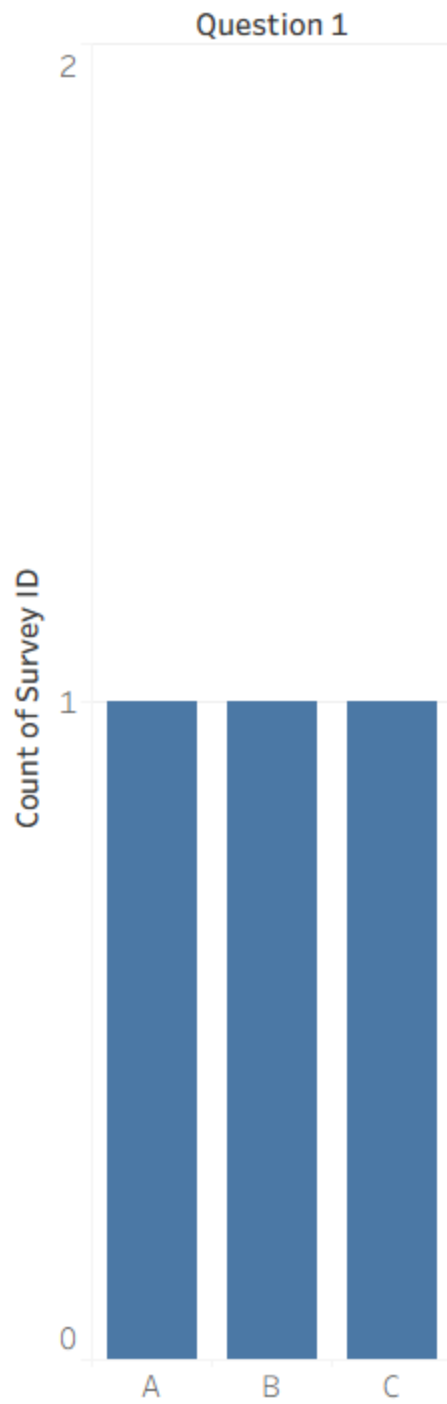
iii Columns

Question 1

Rows

CNT(Survey ID)

Distribution of Question 1 Answers



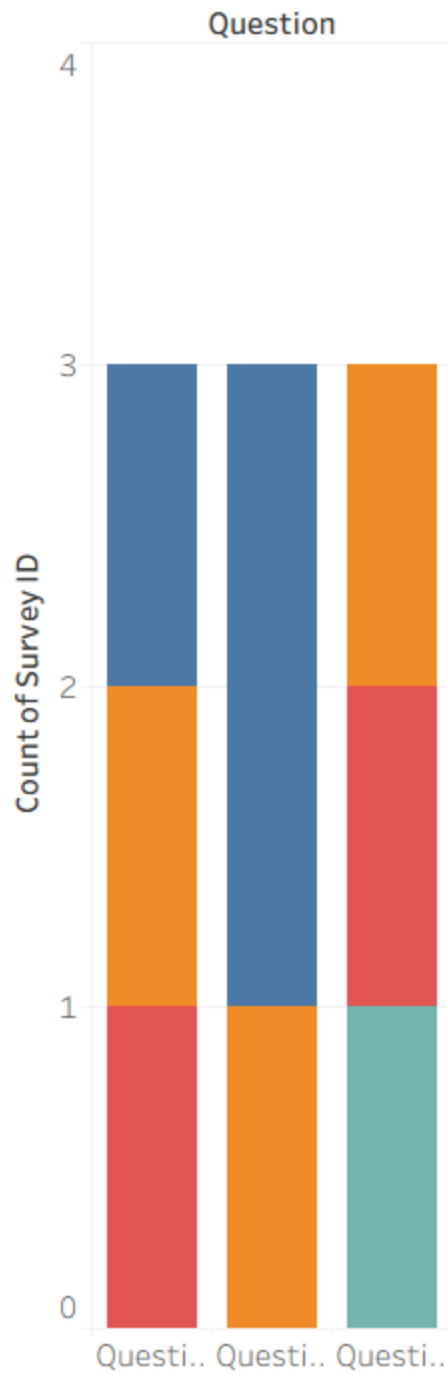
Columns

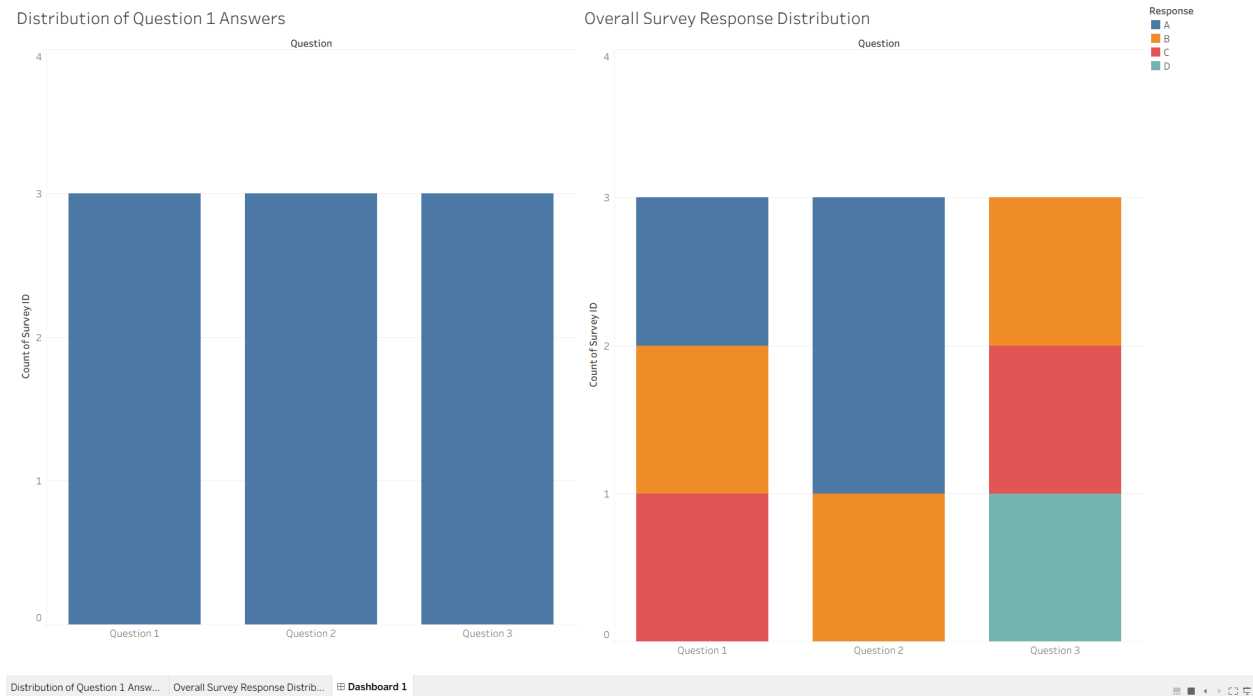
Question

Rows

CNT(Survey ID)

Overall Survey Response Distribution





Procedure

1. Open **RStudio** and enter the given survey response data.
2. Use the `table()` function to calculate the frequency of Question 1 responses.
3. Use `barplot()` with `beside = TRUE` to create the **grouped bar chart**.
4. Count the responses for all three questions and combine them into a response table.
5. Use `barplot()` with `beside = FALSE` to create the **stacked bar chart**.
6. Use `data.frame()` to create a table containing Survey ID and responses to all three questions.
7. Open **Tableau** and import the survey response dataset.
8. Create the grouped bar chart for Question 1.
9. Create the stacked bar chart showing the overall distribution of responses.
10. Create a table displaying the responses for each survey.
11. Combine all three visualizations into a **Tableau dashboard**.
12. Add suitable titles, labels, filters, and formatting for interactive exploration.

Result

The survey response data was successfully analyzed using **R and Tableau**. A grouped bar chart was created to visualize Question 1 responses, a stacked bar chart was created to represent the response distribution across all three questions, and a table was created to display individual survey responses. The three visualizations were successfully combined into an **interactive Tableau dashboard**.